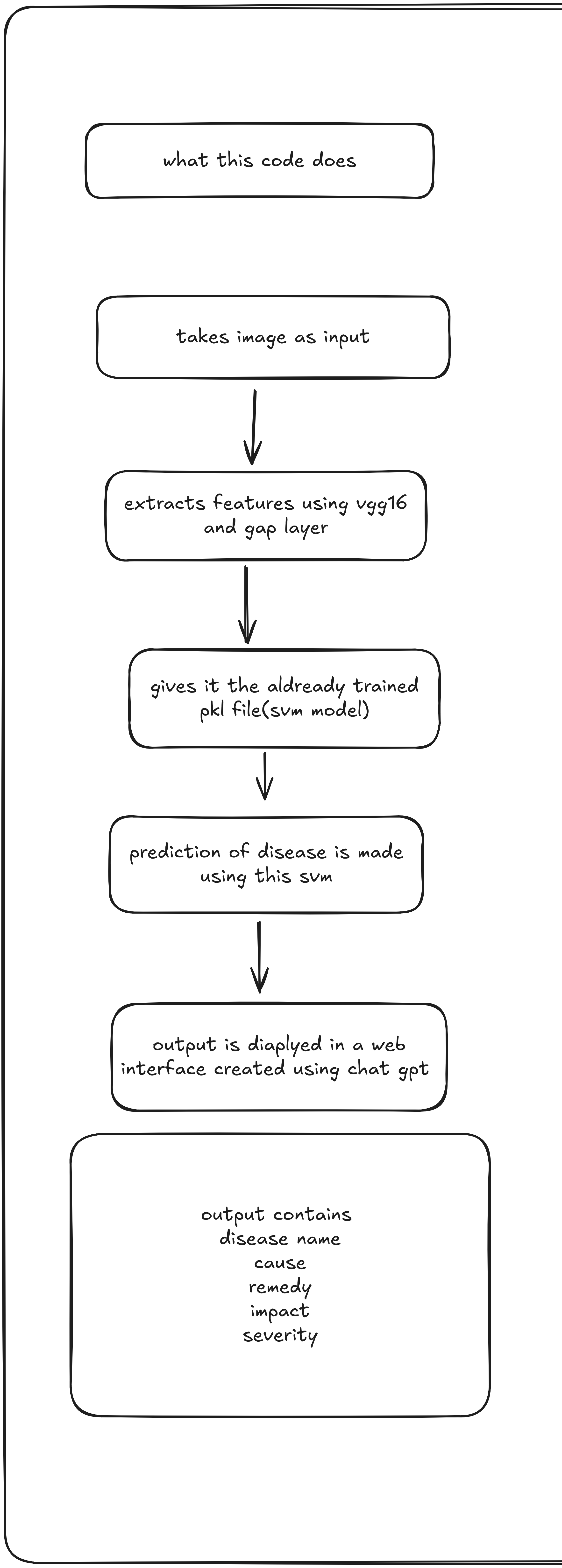

what this code does
takes image as input
extracts features using vgg16 and gap layer
gives it the aldready trained pkl file(svm model)
prediction of disease is made using this svm
output is diaplyed in a web interface created using chat gpt
output contains
disease name
cause
remedy
impact
severity

What is it??!!

Tomato.pkl

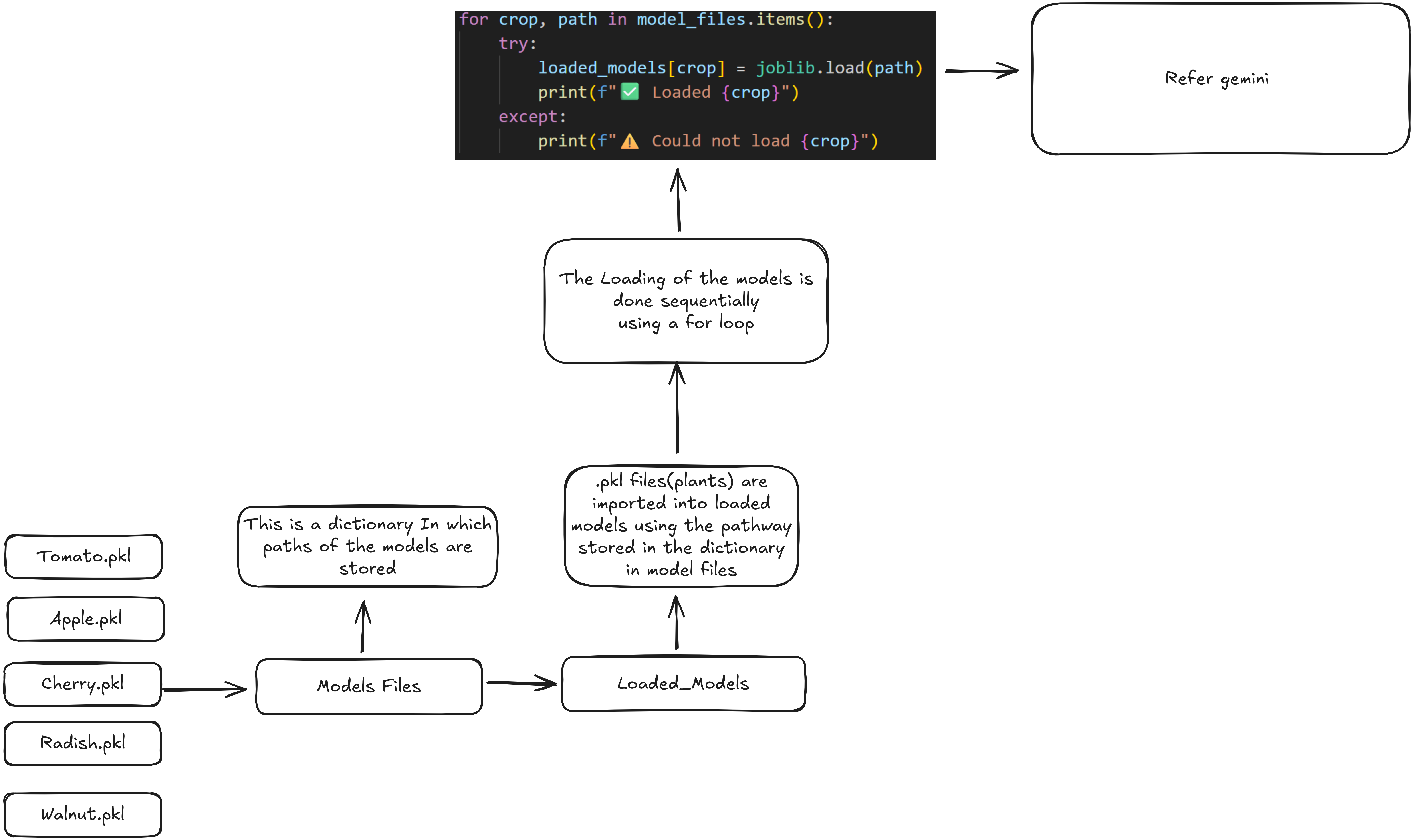
it is a support vector matrice algorithm(machine learning model) which is trained on tomato images for various disease.

it functions in the following manner

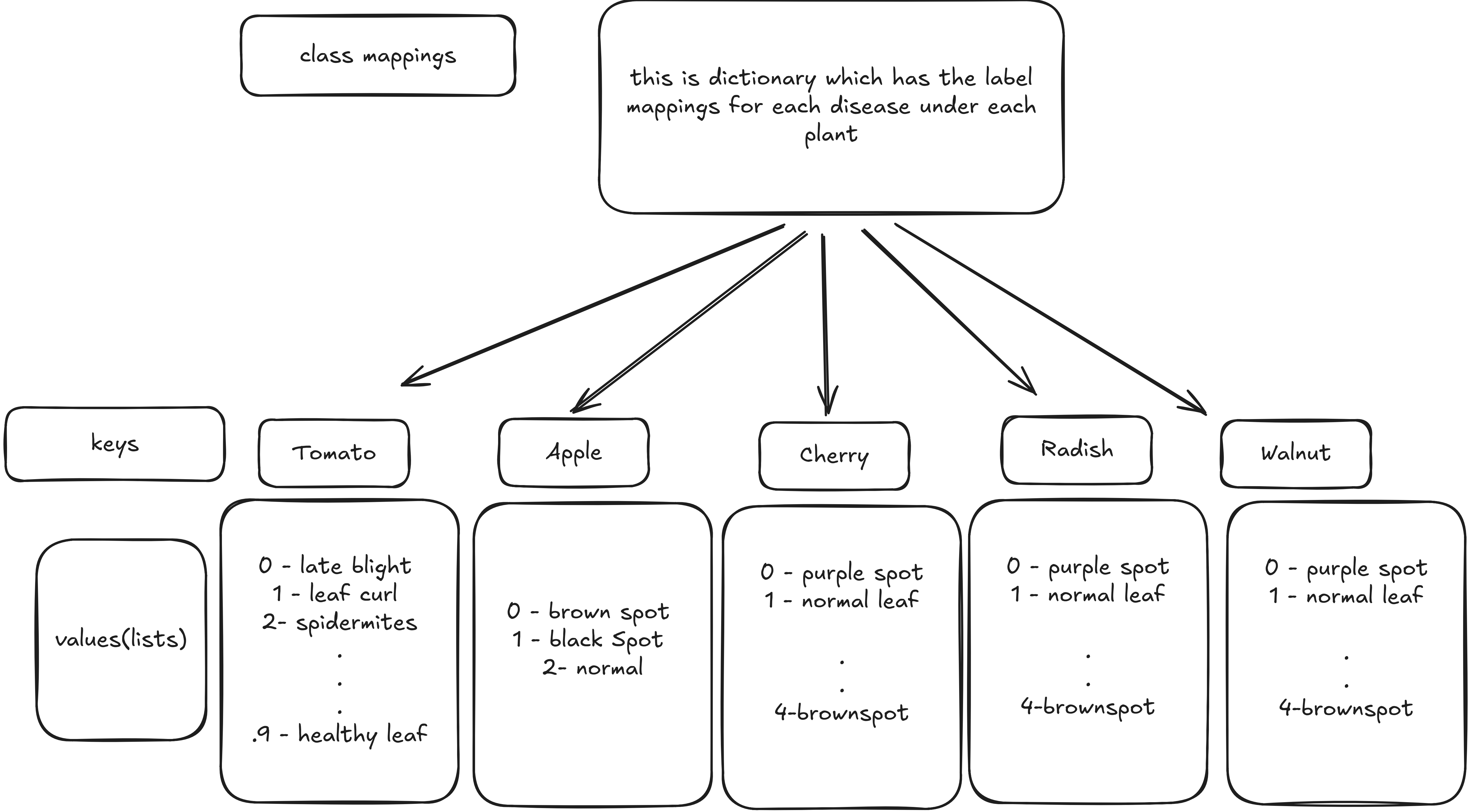
0 - late blight
1 - leaf curl
2- spidermites
.
.
.
.9 - healthy leaf

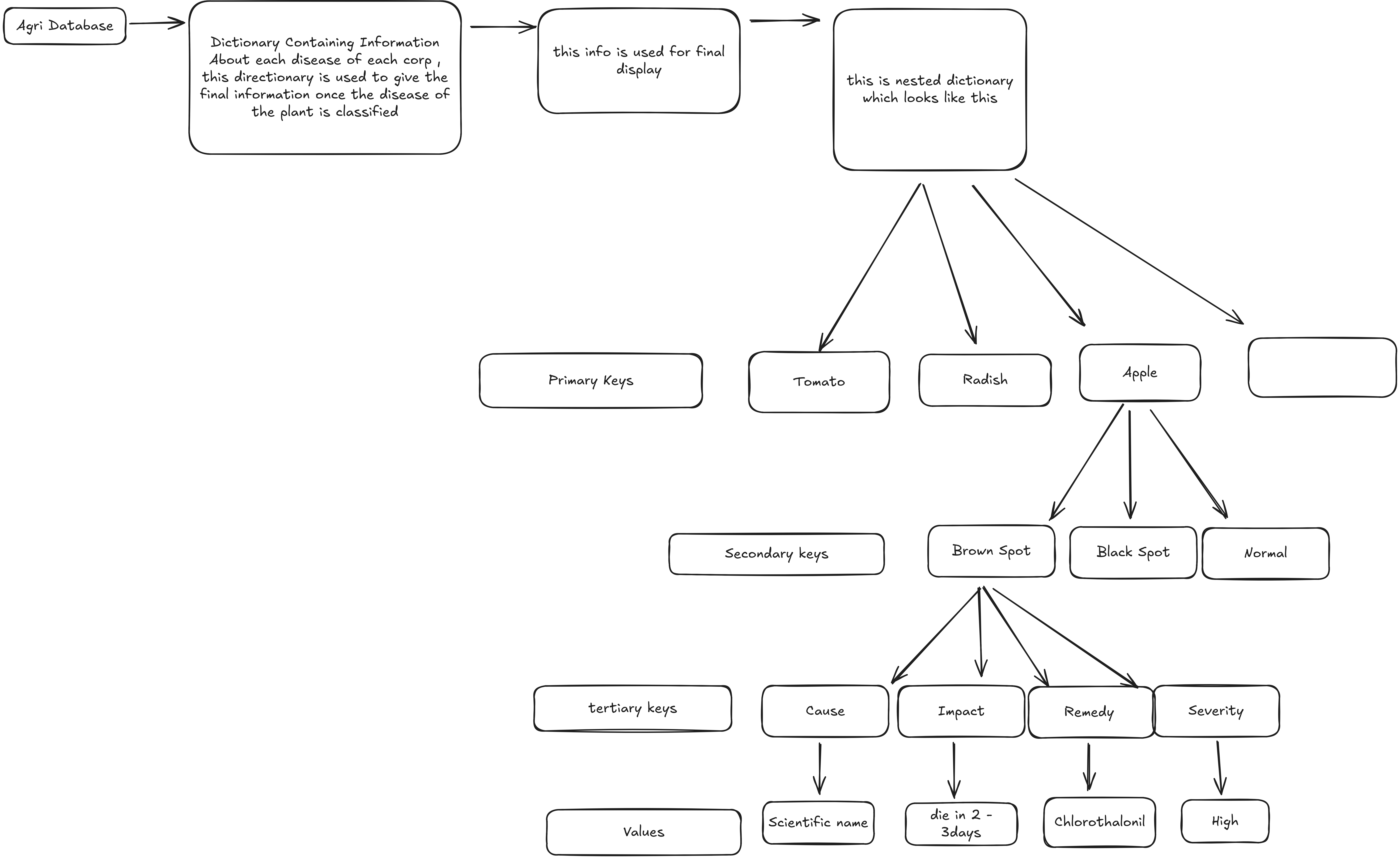
it takes feature vector of an image as an input

it predicts a label ranging from '0 to 9 ' for this given input and gives the label as the output



for crop, path in model_files.items():
try:
loaded_models[crop] = joblib.load(path)
print(f"✅ Loaded {crop}")
except:
print(f"⚠ Could not load {crop}")
Refer gemini
The Loading of the models is done sequentially using a for loop
.pkl files(plants) are imported into loaded models using the pathway stored in the dictionary in model files
This is a dictionary In which paths of the models are stored
Tomato.pkl
Apple.pkl
Cherry.pkl
Radish.pkl
Walnut.pkl
Models Files
Loaded_Models


class mappings
this is dictionary which has the label mappings for each disease under each plant
keys
Tomato
Apple
Cherry
Radish
Walnut
values(lists)
0 - late blight
1 - leaf curl
2- spidermites
.
.
.
.9 - healthy leaf
0 - brown spot
1 - black Spot
2- normal
0 - purple spot
1 - normal leaf
.
.
4-brownspot
0 - purple spot
1 - normal leaf
.
.
4-brownspot
0 - purple spot
1 - normal leaf
.
.
4-brownspot


Agri Database
Dictionary Containing Information About each disease of each corp , this directionary is used to give the final information once the disease of the plant is classified
this info is used for final display
this is nested dictionary which looks like this
Primary Keys
Tomato
Radish
Apple
Secondary keys
Brown Spot
Black Spot
Normal
tertiary keys
Cause
Impact
Remedy
Severity
Values
Scientific name
die in 2 - 3days
Chlorothalonil
High

Prediction function working

Inputs

Name of the crop , image

image

some other format like cmyk
(512 x 1024 x 4)

convert to rgb

Rgb format
(512 x 1024 x 3)

resize image to 224 x 224

Resized image
(224 x 224 x 3)

Convertion into a batch of one

This is done only because tensorflow does not accept single images it only accepts a batch of images
(1 x 224 x 224 x 3)

preprocess input

Preprocessing its applied

extract the features

Vector is made

using the required model for the loaded models dictionary

predict the label of the disease

map the label to the disease

Find the relevant cause,remedy.... from the agri dictionary database

give the final output